



20-Port 10G Uplink Core Routing Switch (NAV-D8R12S)

High-Speed Layer 3 Industrial Connectivity

The NAV-D8R12S is a high-performance L3 managed switch designed for next-generation IP metropolitan area networks and large-scale campus networks. It features 8x 10/100/1000M RJ45 ports and 12x 1/10G SFP+ fiber ports to meet high-density big data transmission requirements.



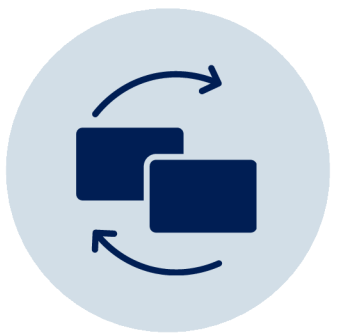
Advanced Hardware

Features a 1U architecture with a switching capacity of up to 680Gbps, ensuring wire-speed forwarding for voice, video, and data triple-play applications.



Robust Security

Supports IEEE 802.1x, Radius, and Tacacs+ authentication. Includes hardware-level deep packet inspection to prevent DoS, TCP SYN Flood, and UDP Flood attacks.



Carrier-Grade Reliability

Supports ISSU (In-Service Software Upgrade) for uninterrupted data forwarding and VRRP/Ring Network protection for millisecond-level fault recovery.

Switching Capacity

680 Gbps

Non-blocking Performance

Product Warranty

3 Years

Lifelong Maintenance

This switch supports advanced routing protocols such as RIP, OSPF, BGP, and PIM-DM/SM, making it ideal for traditional or fully virtualized network environments. Network managers can flexibly choose optical fiber connections to effectively expand 1/10G networks based on transmission distance requirements.

20-Port 10G Uplink Core Routing Switch (NAV-D8R12S)



Product Overview

The **NAV-D8R12S** is a high-performance Layer 3 (L3) managed switch designed as a next-generation 10G convergence solution for IP metropolitan area networks, large-scale campus networks, and enterprise environments. It features 8 x 10/100/1000M RJ45 ports and 12 x 1/10G SFP+ fiber ports.

The NAV-D8R12S offers complete L3 management capabilities with a comprehensive suite of protocols and applications. It provides high-performance L2/L3/L4 wire-speed switching and integrates advanced services such as IPv6 support, MPLS VPN, network security, traffic analysis, and virtualization.

Designed for high reliability, the switch features uninterrupted upgrades, graceful restart, and redundancy protection to ensure continuous network communication. It supports advanced routing protocols like RIP, OSPF, BGP, and PIM-DM/SM, making it ideal for traditional or fully virtualized big data transmission. Network managers can flexibly select optical fiber connections based on transmission distance and speed requirements to effectively expand 1/10G networks.

Key Features

Advanced Hardware Architecture

- 8× Gigabit RJ45 ports + 12× 1/10G SFP+ fiber ports for flexible connectivity.
- 680Gbps switching capacity ensures non-blocking, wire-speed data transmission.

Multi-Layer Security Defense

- Supports IEEE 802.1x, Radius, and Tacacs+ for strict identity authentication.
- Hardware-level deep packet inspection isolates illegal data and blocks DoS/Flood attacks.

Rich L3 Service Capabilities

- Complete L3 routing support (RIP, OSPF, BGP) and MPLS VPN for private networks.
- Comprehensive IPv6 protocol suite and multicast routing (PIM-DM/SM) for video/data convergence.

Carrier-Grade Stability

- ISSU support ensures uninterrupted data forwarding during system upgrades.
- VRRP, Ring Network Protection, and BFD provide millisecond-level fault recovery.

Smart O&M & Efficiency

- Diverse management via Web, CLI, SNMP (V1/V2/V3), and RMON for network optimization.
- IEEE 802.3az energy-saving design with intelligent, speed-regulating fans.

A high-performance L3 core routing switch with 12× 10G SFP+ uplinks and 8× Gigabit ports, designed for large campus and enterprise networks. It offers strong virtualization, security, and "slow start" power protection for enhanced reliability.

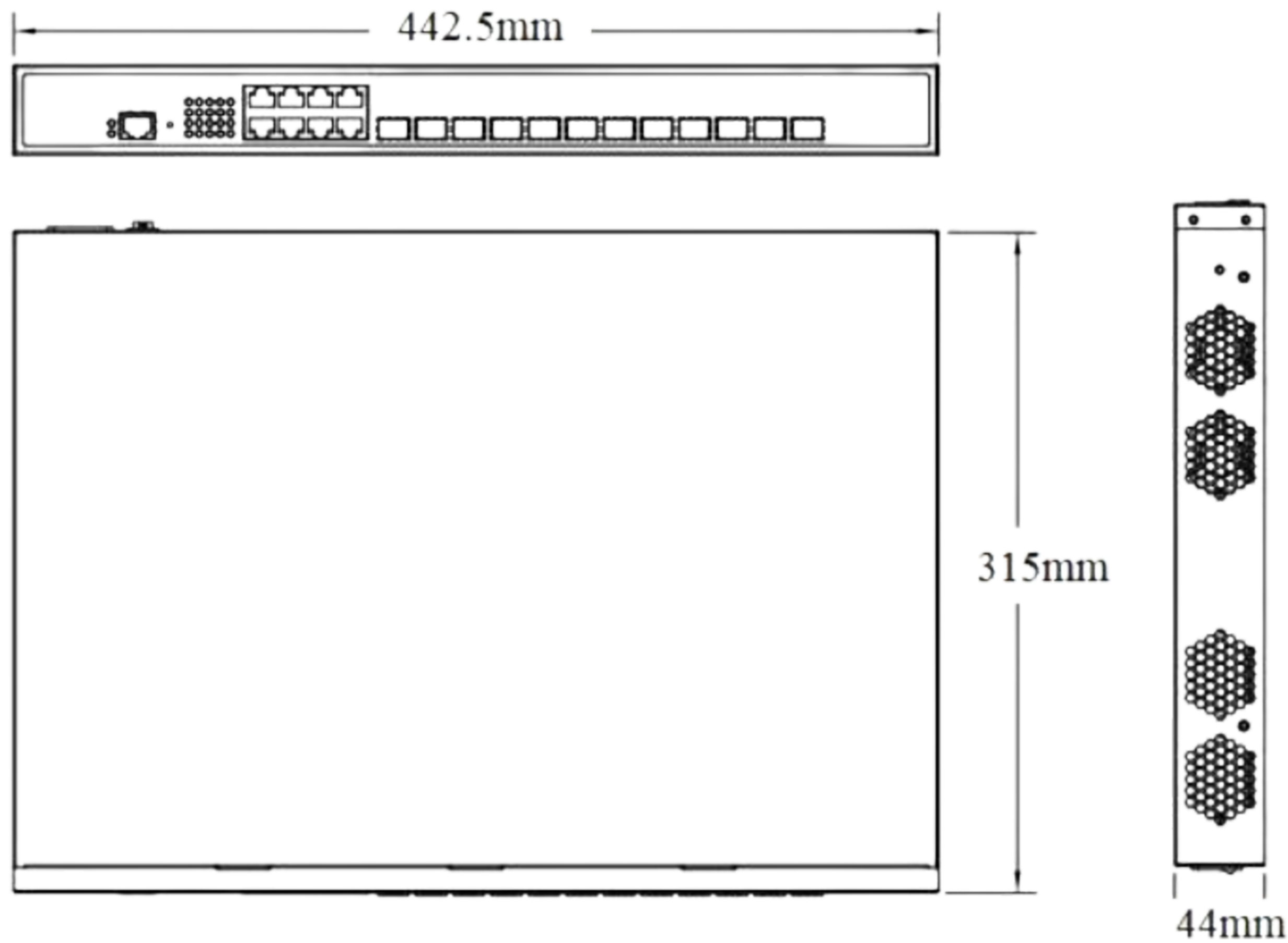
Technical Specification

Feature	Specification
Model	NAV-D8R12S
Interface Characteristics	12*1/10G SFP+ fiber ports (Data)
Fixed Port	8*10/100/1000M RJ45 ports (Data) 1*Console RS232 port (9600,8,N,1)
Ethernet Port	Port 1-8 can support 10/100/1000Base-T auto-sensing, full/ half duplex MDI/MDI-X self-adaption
	10BASE-T: Cat3,4,5 UTP(≤100 meters)
Twisted Pair Transmission	100BASE-TX: Cat5 or later UTP(≤100 meters) 1000BASE-T: Cat5e or later UTP(≤100 meters)
Optical Fiber Port	1/10G SFP+ optical fiber interface, default no include optical modules (optional single-mode/ multi-mode, single fiber/ dual fiber optical module. LC)
Optical Cable/ Distance	Multi-mode: 850nm/0-550m, Single-mode: 1310nm/0-40km, 1550nm/ 0-120km.
Chip Parameter	
Network Management Type	L3
Network Protocol	IEEE 802.3u 100Base-TX, IEEE 802.3ab 1000Base-T, IEEE 802.3z 1000Base-X, IEEE 802.3ae 10Gb/s Ethernet, IEEE802.3x
Forwarding Mode	Store and Forward (Full Wire Speed)
Switching Capacity	680Gbps (non-blocking)
Forwarding Rate@64byte	136Mpps
MAC	32K
Buffer Memory	32M
Jumbo Frame	9K
LED Indicator	Power: PWR (Green), System: SYS (Green), Network: G1-G8 (Green), Fiber: TE1-12 (Green)
Reset Switch	Yes. Press and hold for 10 seconds and release to restore factory settings
Power Supply	
Total PWR/ Input Voltage	75W/ (AC100-240V)
Power Consumption	Standby<20W, Full Load<40W
Power Supply	Built-in power supply, AC100~240V 50-60Hz, 1.0A
Physical Parameter	
Operation Temp/Humidity	-20°C~+55°C, 5%~90% RH Non condensing
Storage Temp/ Humidity	-40°C~+75°C 5%~95% RH Non condensing
Dimension (L*W*H)	442.50*315*44mm
Net/Gross Weight	<5.8kg/<6.2kg
Installation	Desktop, 1U/19 inch cabinet

Certification & Warranty	
Lightning Protection	Port lightning protection: 6KV 8/20us, Protection level: IP30
Certification	CCC, CE mark, commercial, CE/LVD EN62368-1, FCC Part 15 Class B, RoHS
Warranty	3 years, lifelong maintenance.
Network Management Feature	
Virtualization and Stacking	VRRP virtualization stack, active/standby switching Policy routing, BFD for OSPF, BGP
IPv4	Equal-cost routing to achieve load balancing Static routing, RIP v1/v2, OSPF, BGP, IS-IS, BEIGRP MLD v1/v2, IPv6 neighbor discovery MLD Snooping, Path MTU discovery
IPv6	ICMPv6, DHCPv6, ACLv6, IPv6 Telnet IPv6 static routing, RIPng, OSPFv3, BGP4+ Manual tunnel, ISATAP tunnel, 6to4 tunnel
MAC Switching	Configuring MAC address aging time IEEE 802.1AE MacSec security control Static configuration and dynamically learning of MAC address MAC address filtering function, Check and delete MAC address Black-hole MAC table entries, Limit on MAC address learning number
VLAN	Basic QinQ and flexible QinQ function GVRP, Private VLAN, 4K Active VLAN, 1:1 and N:1 VLAN Mapping
Link Aggregation	10GE port aggregation, Static aggregation, Dynamic aggregation
Flow Monitoring	SFLOW, DHCP Snooping option82/ DHCP Relay option82
DHCP	DHCP Client, DHCP Snooping, DHCP Relay, DHCP Server Zero configuration methods such as DHCP auto-config and CWMP-TR069
STP/ERPS	802.1D (STP), 802.1W (RSTP), 802.1S (MSTP) BPDU protection, Root protection, Loop protection ERPS Ethernet Ring Protection Protocol (G.8032)
Multicast	Multicast group policy and multicast number limit IGMP V1/v2/v3, IGMP Snooping, IGMP Fast Leave PIM-SM, PIM-DM, Multicast traffic cross VLAN duplication
ARP	Static entries, ARP source suppression Standard proxy ARP and local proxy ARP Free ARP, Dynamic ARP Inspection, ARP anti-attack ARP Detection (check based on DHCP Snooping security entries, 802.1x entries, or IP/MAC static binding entries)

Mirroring	Flow mirroring, N:4 port mirroring, Local and remote port mirroring
MPLS VPN	MCE, MPLS TE, MPLS OAM, LDP protocol, P/PE of MPLS VPN
QoS/ACL	Traffic supervision and traffic shaping SP, WRR, SP+WRR scheduling mode CAR traffic restriction, 802.1P/DSCP priority re-marking Tail-Drop, WRED, and other congestion avoidance mechanisms Traffic classification based on each field of the L2/L3/L4 protocol header Ingress and Egress ACL, matching L2, L3, L4 and IP five-tuple, copying, forwarding, and discarding Hash same-source and same-destination load balancing to ensure session integrity of traffic output Identification and filtering of L2/L3/L4 based ACL
Security	Urpf, Port isolation, Port security, IP + MAC + port binding IEEE 802.1x certification, DHCP Snooping, DHCP Option 82 Radius and BDTacacs+, Command line hierarchical protection Suppression of broadcast, multicast, and unknown unicast packet Defend against DDoS attack, SYN Flood attack of TCP, and UDP Flood attack
Reliability	EAPS, ERPS ring network protection ISSU service without interruption system upgrade HSRP, VRRP hot standby protocol, GR for OSPF, BGP Optional power supply 1+1 backup, BFD for OSPF, BGP Static/LACP link aggregation, support cross-service card link aggregation
Management	File upload and download management in TFTP mode Telnet remote maintenance, ZTP (Zero Touch Provisioning) Power alarm, Fan, temperature alarm, Console, Telnet, SSH 2.0 ISSU, Track, Tracert, SFLOW and other traffic statistics analysis NTP, Ping, Debug information output, Web browser management System logs, Graded alarm, SNMP v1/v2/v3, 802.1AG and 802.3AH RMON event history, SNMP (Simple Network Management Protocol)
Energy Saving	IEEE802.3az green energy Ethernet
System	Web browser: Mozilla Firefox 2.5 or higher, Google Chrome V42 or higher, Cat5 and above Ethernet cable TCP/IP, network adapter, and network operating system (such as Microsoft Windows, Linux, Mac OS X) installed on each computer in the network

Dimensions



Ordering Information

Model	Description	Built-in Power Supply
NAV-D8R12S	L3 managed 10G uplink Ethernet core routing switch with 810/100/1000M RJ45 ports and 121/10G SFP+ fiber ports. Built-in power supply and supports 1U/19 inch cabinet installation.	75W
Note: The optical module is not included and needs to be purchased.		

Packing List

Content	Qty	Unit
20-port 10G uplink core routing switch	1	Set
AC Power Cable	1	PC
RJ45-DB9 Adapter Cable	1	PC
Mounting Kit	1	Set
Warranty Card and Certificate of Conformity	1	PC

Optical Modules

Product	Model	Description	Unit
1.25G Optical Module	2630	SFP optical module, 1.25G multi-mode dual fiber 850nm, transmission distance: 550m, LC interface. supports DDM function and hot plugging.	PC
	2632	SFP optical module, 1.25G single-mode dual fiber 1310nm, transmission distance: 20km, LC interface. supports DDM function and hot plugging.	PC
	2612-T	SFP optical module, 1.25G single-mode single fiber TX1310nm/ RX1550nm, transmission distance: 20km, LC interface. supports DDM function and hot plugging.	PC
	2613-R	SFP optical module, 1.25G single-mode single fiber TX1550nm/ RX1310nm, transmission distance: 20km, LC interface. supports DDM function and hot plugging.	PC
	2612-T-SC	SFP optical module, 1.25G single-mode single fiber TX1310nm/ RX1550nm, transmission distance: 20km, SC interface. supports DDM function and hot plugging.	PC
	2613-R-SC	SFP optical module, 1.25G single-mode single fiber TX1550nm/ RX1310nm, transmission distance: 20km, SC interface. supports DDM function and hot plugging.	PC
Power Module	2633	1.25G SFP optical module transfers to 10/100/1000M RJ45 port.	PC
10G Optical Module	6630	SFP+ optical module, 10G multi-mode dual fiber 850nm, transmission distance: 300m, LC interface. supports DDM function and hot plugging.	PC
	7832	SFP+ optical module, 10G single-mode dual fiber 1310nm, transmission distance: 20km, LC interface, supports DDM function and hot plugging.	PC
	7832-33	SFP+ optical module, 10G single-mode single fiber TX1330nm/ RX1270nm, transmission distance: 20km, LC interface. supports DDM function and hot plugging.	PC
	7832-27	SFP+ optical module, 10G single-mode single fiber TX1270nm/ RX13300nm, transmission distance: 20km, LC interface. supports DDM function and hot plugging.	PC